

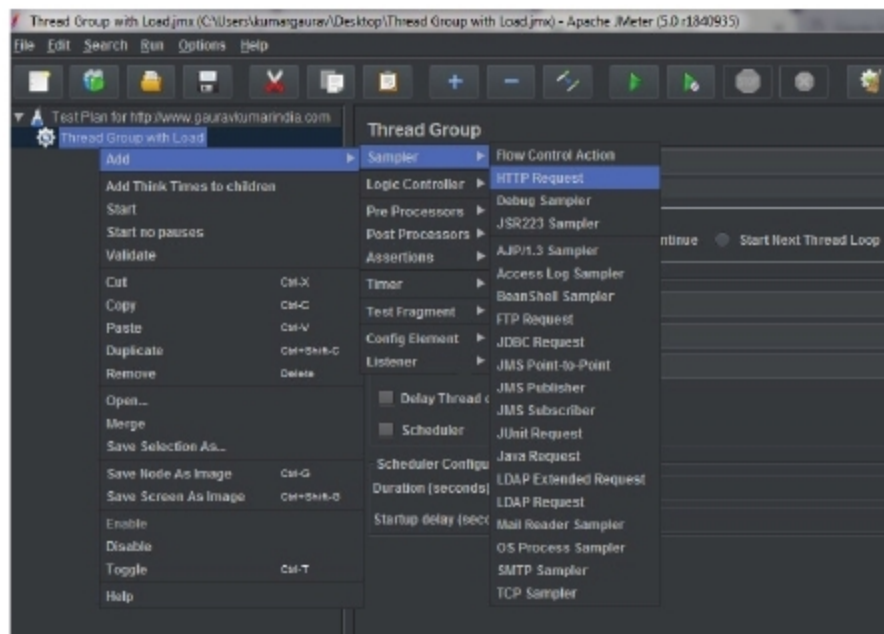


Figure 8: Generating HTTP requests to simulate load

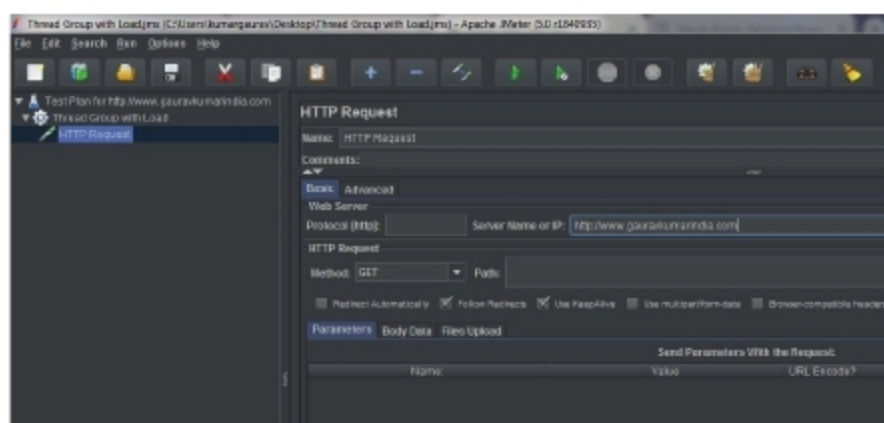


Figure 9: Specifying the Web application or IP address for load testing

and factors are successful under assorted and large loads on the Web application (Figure 10).

From the analysis of logs and the response from the server, it can be checked whether the request to the Web application was successful or not (Figure 11). The response code presents the cumulative outcome from the server, right from the initial point of contact by the client.

HTTP Unbearable Load King (HULK) automation for load testing

Generally, penetration testers are dependent on third party tools and software applications for digital forensics and to audit Web applications. This approach of using third party software is not always reliable and secure.

Different testing strategies can be implemented using core programming scripts written in Python. Even though there are many tools and libraries for load testing, stress testing and performance testing, Python programs can be directly used for these tests. Using unreliable tools for penetration and load testing can be dangerous because our own system's information can be shared to the other locations and will not be secure.

HTTP Unbearable Load King (HULK) is a type of DDoS (distributed denial of service) attack on the server but it can be used for testing Web applications too. By implementing HULK, the performance of a Web application can be tested to check whether it is able

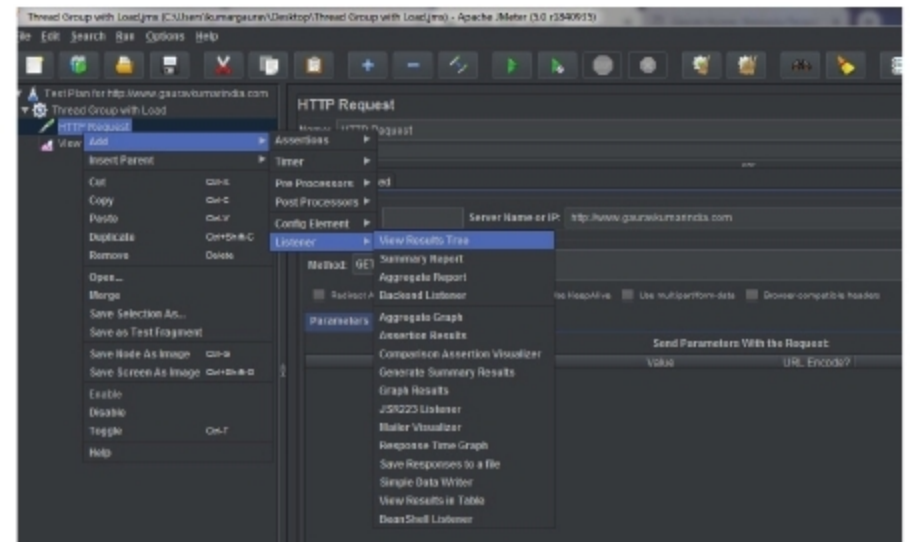


Figure 10: Integration of the 'Results Tree'

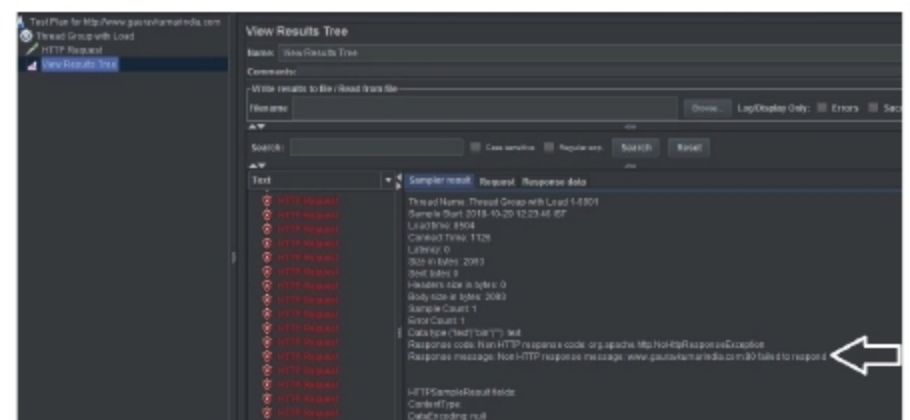


Figure 11: Analysis of logs and the response from the server

to handle the DDoS or HULK based traffic. If the Web application goes down or faces excessive delay in opening, preventive measures can be taken. The HULK script generates an unbearably huge load on the Web application, helping the Web administrator to check whether the application will work fine in real-world scenarios.

Web masters and network administrators can analyse the error log files and access the log files to check the DDoS attempts, and also evaluate the performance of a Web application after carrying out HULK or similar implementations.

Scope for R&D

In the domain of security for Web based applications, there are many perspectives that need to be addressed and worked out. The following are some of the aspects and research points that can be solved by researchers, corporate organisations and academicians in the field of software testing and forensics based audits:

- Biometric integrated access for secured Web portals;
- Development of trust architecture for secured Web applications;
- Integrity-aware vulnerability recognition in Web applications; and
- Deep learning based identification of suspects accessing Web applications without permissions. **END** 🐧

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